3.3. Recurrence properties.

Def. cpt $A \subset X$: attractor for f if $\exists$ nbd V of A & $N \in \mathbb{N}$ s.t. $f^N(V) \subset V$ & $A = \bigcap f^n(V)$.

$y \in X$: ω-limit pt (resp. α-limit pt) for $x \in X$ if $\exists$ seq. of moments of time going to ∞ (resp. $-\infty$) s.t. images of x converge to y.

$$\omega(x) = \bigcap_{T=0}^{\infty} \overline{\left(\bigcup_{t \geq T} \varphi^t(x) \right)}, \qquad \alpha(x) = \bigcap_{T=0}^{-\infty} \overline{\left(\bigcup_{t \leq T} \varphi^t(x) \right)}$$

e.g. $f: S^1 \to S^1$, $x \mapsto x + \frac{1}{10} \sin^2 \pi x \pmod 1$

$x=0$: Not attractor, but $\alpha(0) = \omega(0) = \{0\}$.

Def. $x \in X$: positively recurrent (resp. negatively recurrent) if $x \in \omega(x)$ (resp. $x \in \alpha(x)$)

recurrent if $x \in \alpha(x) \cap \omega(x)$.

$R^+(f)$, $R^-(f)$, $R(f)$: closure of pos. / neg. / - recurrent pts.

Def. $x \in X$: nonwandering w.r.t. $f: X \to X$ if $\forall$ open $U \ni x$, $\exists N > 0$ s.t. $f^N(U) \cap U \neq \phi$.

$NW(f)$

Rmk. $\exists N$ s.t. $n \geq N \Rightarrow f^n(U) \cap U = \phi$ $\forall$ open $U \ni x$

$\Rightarrow$ x : non-wandering ($\because$ x is not a periodic pt.)

Prop. $NW(f)$: closed & f-inv.

contains $\forall$ ω-, α-limit pts for all pts.

Cor. $f: X \to X$ (X cpt) $\Rightarrow$ $NW(f) \neq \phi$

$NW_1(f) = NW(f)$, ..., $NW_{n+1}(f) = NW(f|_{NW_n(f)})$

$NW_\omega(f) = \bigcap NW_i(f)$

↳ again ... $\Rightarrow NW_{\omega+1}(f) = NW(f|_{NW_\omega(f)})$

↳ transfinite induction stops at a countable ordinal.

↳ "center" of the dynamical system.

$NW(f)$ or $NW_2(f)$ in many cases.

$NW(f)$: "hub" of recurrence behavior.

α- / ω- limits, recurrence pts, periodic pts.

$$h_{top}(f) = h_{top}(f|_{NW(f)}).$$

e.g. $f_\lambda : \mathbb{C} \to \mathbb{C}$, $z \mapsto (1-\lambda) z^2$ $\quad (\lambda \in [0,1])$

- $\lambda = 0$; $NW(f_0) = S^1 \cup \{0\}$. $\rightsquigarrow h_{top} = \log 2$
- $\lambda > 0$; $NW(f_\lambda) = \{0\}$. $\rightsquigarrow h_{top} = 0$.

Prop. conti. f on cpt metric sp. X has an inv. minimal set.

Cor. $R(f) \neq \phi$.

$$\mathrm{Per}(f) \subset M(f) \subset R(f) \subset R^+(f) \cup R^-(f) \subset NW(f).$$

$M(f)$ = closure of union of inv. minimal sets.

nontrivial recurrence : $\exists$ nonperiodic recurrent pts.

↳ low dim dynamics
- homeo of circle Ch 11
- flows on surfaces Ch 14

Def. $f: X \to X$: **topological transitive**

if $\exists\ x \in X$ s.t. $\mathcal{O}_f(x) = \{f^n(x)\}_{n\in\mathbb{Z}}$: dense

Def. $f: X \to X$: **minimal**

if $\mathcal{O}_f(x)$ is dense $\forall x \in X$.

equiv. to. f has no proper closed inv. sets.

Def. $f: X \to X$: **topological mixing**.

if $\forall$ open $U, V \subset X$, $\exists\ N > 0$ s.t.

$n \geq N \Rightarrow f^n(U) \cap V \neq \phi$.

& recurrence, ...

$\Downarrow$

Asymptotic frequencies "measure theory"

typical behavior w.r.t. an inv. measure.

$$\begin{cases} \text{pt} & \to \quad \text{recurrence} \\ \mu\text{-a.e.} & \to \quad \text{ergodic} \\ \exists!\,\mu & \to \quad \text{uniquely ergodic} \end{cases}$$

4.1. Asymptotic distribution & statistical behavior of orbits.

f : contz. self map of metrizable sp. X.

$x \in X$, "sufficiently good" set $U \subset X$.

$F_U(f, x, n) :=$ # of integer $k = 0, \dots, n-1$ s.t. $f^k(x) \in U$.

$$\Rightarrow F_U(f, x) \overset{\exists ?}{:=} \lim_{n\to\infty} \frac{F_U(f, x, n)}{n} \ : \text{asymptotic density of the distribution}$$

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \chi_U(f^k(x)) . \ : \text{time average / Birkhoff –}$$

of the char. ft. χ_U.

f : fixed.

contz.

$$I_x(\varphi) := \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \varphi(f^k(x))$$

$I_x : C(X) \to \mathbb{R}$ satisfies

- linearity
- bdd : $|I_x(\varphi)| \le \sup |\varphi(y)|$ $\} \to I_x$: contz.
- positivity : $I_x\varphi \ge 0$ if $\varphi \ge 0$, $I_x 1 = 1$.
- inv. under f : $I_x(\varphi \circ f) = I_x(\varphi)$ $(= I_{f(x)}(\varphi))$.

$\hookrightarrow \mu(f^{-1}A) = \mu(A)$

Riesz rep. Thm.

pos. bdd. action functional $\Rightarrow \exists!$ Borel probability measure μ.

i.e. $I_x(\varphi) = \int_X \varphi \, d\mu$.

Main questions :

- $\exists ? \ x \in X$ s.t. asymptotic distributions I_x exists.
- f-invariant measure $\mu \Rightarrow$ Can we "find" such $x \in X$?

Thm. (Krylov - Bogolubov).

Any contr. on a cpt metrizable sp. X has an inv. BP measure.

sketch) $x \in X$, $\{\varphi_1, \varphi_2, \cdots\}$: Countable dense in $C(X)$.

For each m, $\frac{1}{n}\sum_{k=0}^{n-1} \varphi_m(f^k(x))$: bdd.

$\exists$ conv. subseq $\Rightarrow$ diagonal process $\Rightarrow$ μ_x by RRT.

Thm (Birkhoff Ergodic Theorem).

$T : (X,\mu) \to (X,\mu)$: measure preserving on probability sp. & $\varphi \in L^1(X,\mu)$

For μ-almost every $x \in X$, time average exists ;

$$\varphi_T(x) := \lim_{n\to\infty} \sum_{k=0}^{n-1} \varphi(T^k(x)) .$$

sketch) · $\mathcal{I}$: T-inv. σ-algebra.

· $\varphi \in L^1 \Rightarrow \varphi_{\mathcal{I}} = \dfrac{d(\varphi\mu)|_{\mathcal{I}}}{d\mu|_{\mathcal{I}}}$: Radon-Nikodym derivative

· $F_n := \max_{k\le n} \sum_{i=0}^{k-1} f\circ T^i$. $f = \varphi - \varphi_{\mathcal{I}} \in L^1$

LDCT $\downarrow$

$\varphi_T = \varphi_{\mathcal{I}}$ μ-a.e.

Rmk. T : invertible $\Rightarrow$ backword time average

two-sided time avarage $\frac{1}{2n-1}\sum_{-n+1}^{n-1} \varphi(T^k(x))$

Prop. $\varphi_T(x) = \overline{\varphi}_T(x)$ $(:= \varphi_{T^{-1}}(x))$ a.e.

Cor. X: cpt metrizable, f: conti.

$\{x : \exists\, I_x(\varphi)\ \forall$ conti. $\varphi\}$: full measure for any f-inv. BP measure.

Cor. f: conti. on cpt metric sp. X.

$\Rightarrow \exists\, x \in X$ s.t. $I_x(\varphi)$ is well-defined $\forall$ conti. φ.

Def. f-inv. measure μ : ergodic w.r.t. f

if $\mu(A) = 0$ or $\mu(X \setminus A) = 0$ $\quad \forall$ f-inv. $A \subset X$.

conti. f : uniquely ergodic if it has only one inv. BP —

Prop. The only inv. BP measure is ergodic.

proof) $0 < \mu(A) < 1 \Rightarrow \mu_A(B) := \dfrac{\mu(A \cap B)}{\mu(A)}$

$\mu_A, \mu_{X \setminus A}$: different BP measures $\divideontimes$

Cor. f: ergodic μ-preserving, $\mu(X) = 1$, $\varphi \in L^1$

$$\Rightarrow \varphi_f(x) = \lim_{n \to \infty} \frac{1}{n} \sum_{k=0}^{n-1} \varphi(f^k(x)) = \int_X \varphi\, d\mu \quad \forall x \text{ outside a measure 0 set}$$

f-inv. $\Rightarrow$ const a.e. (Birkhoff) $\varphi_f = \varphi_I$ μ-a.e.

$\mathcal{M}(f) := \{f\text{-inv. BP measures.}\}$: ∞-dim.

extreme pts of $\mathcal{M}(f)$: ergodic measures

$v = ta + (1-t)b \Rightarrow a = b = v$ or $t = 0$ or 1 in convex set

Rmk. · μ : invariant BP measure $\Rightarrow \mu = \sum \mu_\alpha$ ergodic.

· f: uniquely ergodic $\Rightarrow \forall$ conti. φ, $\frac{1}{n} \sum \varphi(f^k(x))$ conv. unif.

$\Leftarrow$ holds if f is topologically transitive.

· U: open, $\mu(\partial U) = 0 \Rightarrow \frac{1}{n} \sum_{k=0}^{n-1} \chi_U(f^k(x)) \to \mu(U)$ unif.

Def. μ: Borel measure on a separable metrizable sp. X.

$\operatorname{supp}\mu := \{x\in X : \mu(U)>0 \ \forall \text{ open } U \text{ containing } x\}$: support of μ.

Prop. · $\operatorname{supp}\mu$: closed

· $\mu(X\setminus \operatorname{supp}\mu)=0$

· $\forall$ full measure set : dense in $\operatorname{supp}\mu$.

Thm (Poincaré Recurrence Theorem).

T : measure preserving transf. of P sp. (X,μ), $A\subset X$: measurable

$$\mu(\{x\in A : \{T^n(x)\}_{n\geq N}\subset X\setminus A\})=0 \quad \forall N\in\mathbb{N}.$$

sketch) $\tilde{A} := A\cap\left(\bigcap T^{-n}(X\setminus A)\right) \Rightarrow \tilde{A}\cap T^{-n}(\tilde{A})=\phi$

$$T^{-n}(\tilde{A})\cap T^{-m}(\tilde{A})=\phi \quad \& \ \mu(\tilde{A})=\mu(T^{-n}(\tilde{A}))$$

$$1=\mu(X)\geq \mu\left(\bigcup T^{-n}(\tilde{A})\right)=\sum \mu(T^{-n}(\tilde{A}))=\sum\mu(\tilde{A}).$$

Prop. f: conti. on a complete separable metrizable X.

(✱) · $\operatorname{supp}\mu\subset R(f)$ $\forall$ f-inv. BP measure μ.

· μ: ergodic $\Rightarrow$ $\exists$ dense orbit for $f|_{\operatorname{supp}\mu}$. (topological transitivity)

· X: cpt, $f|_{\operatorname{supp}\mu}$: uniquely ergodic $\Rightarrow$ $\operatorname{supp}\mu$ is a minimal set.

sketch) · PRT on countable base $\{U_i\}$

$\Rightarrow \exists$ full measure set $\bigcup \tilde{U}_i$ $\subset R(f)$: dense in $\operatorname{supp}\mu$,

· countable base for $\operatorname{supp}\mu$ $\{U_i\}$ $\Rightarrow$ $\mu(U_i)>0$.

$$\chi_{U_i} \rightsquigarrow \lim \frac{1}{n}\chi_{U_i}(f^n(x)) = \int_X \chi_{U_i}\, d\mu = \mu(U_i)>0. \quad (x\in \underset{\text{full measure}}{R_i})$$

· $\exists$ proper closed inv. subset $\Lambda\subset\operatorname{supp}\mu$ $\underset{\text{(Krylov-Bogolubov)}}{\Rightarrow}$ $\exists$ inv. measure for $f|_\Lambda$

4.2. Examples of ergodicity ; mixing.

rotation / expanding map / hyperbolic toral automorphism.

* Rotations

Prop. (Kronecker-Weyl Equidistribution Theorem).

Any irrational rotation on circle is uniquely ergodic.

$$\Rightarrow \frac{1}{n}\sum_{k=0}^{n-1}\varphi(R_\alpha^k x) \longrightarrow \int_{S^1}\varphi\, d\theta.$$

sketch) $\{\chi_m(x) = e^{2\pi i m x}\}$

$$m\neq 0, \quad \chi_m(R_\alpha x) = e^{2\pi i m(x+\alpha)} = e^{2\pi i m\alpha}\cdot\chi_m(x)$$

$$\left|\frac{1}{n}\sum_{k=0}^{n-1}\chi_m(R_\alpha^k x)\right| = \frac{1}{n}\left|\sum_{k=0}^{n-1} e^{2\pi i m k\alpha}\right| = \frac{|1-e^{2\pi i m n\alpha}|}{n|1-e^{2\pi i m\alpha}|} \le \frac{2}{n\cdot c} \to 0$$

· $T_\gamma : \mathbb{T}^n \to \mathbb{T}^n$, $(x_1,\dots,x_n)\mapsto(x_1+\gamma_1,\dots,x_n+\gamma_n)$

· $T_\omega^t : \mathbb{T}^n \to \mathbb{T}^n$: $(x_1,\dots,x_n)\mapsto(x_1+t\omega_1,\dots,x_n+t\omega_n)$.

Prop. $\gamma_1,\dots,\gamma_n, 1$: rationally indep. $\Rightarrow$ T_γ : uniquely ergodic w.r.t. Lebesgue measure.

sketch) $$\chi(x_1,\dots,x_n) = \sum_{k_1,\dots,k_n\in\mathbb{Z}} \chi_{k_1,\dots,k_n}\exp(2\pi i\textstyle\sum k_j x_j).$$

$$\Rightarrow \quad \chi_{k_1,\dots,k_n}(1-\exp(2\pi i\textstyle\sum k_j \gamma_j)) = 0$$

$$\Rightarrow \quad \chi_{k_1,\dots,k_n} = 0 \quad \text{except } k_1 = \dots = k_n = 0.$$

* expanding maps

NOT uniquely.

Prop. E_m $(|m|\geq 2)$: ergodic w.r.t. Lebesgue measure.

sketch) $A\subset S^1$: measurable E_m-inv. set, positive measure. $\exists$ if $\mu(A)<1$.

$\varepsilon>0$. Δ : interval of length $|m|^{-n}$ s.t. $\lambda(\Delta\setminus A) > (1-\varepsilon)\lambda(\Delta) = (1-\varepsilon)|m|^{-n}$

$$\lambda(E_m^n(\Delta)\setminus A) = \lambda(E_m^n(\Delta\setminus A)) = |m|^n\lambda(\Delta\setminus A) > 1-\varepsilon.$$

* mixing.

Def. measure preserving transformation $T:(X,\mu)\to(X,\mu)$: mixing.

if $\mu(T^{-n}(A)\cap B)\to\mu(A)\cdot\mu(B)$ $\forall$ measurable A, B.

Rmk. mixing $\Rightarrow$ ergodic $(\because \mu(T^{-n}(A)\cap(X\setminus A))=0)$

Prop. · T_γ : NOT mixing.

· E_m $(|m|\geq 2)$: mixing.

proof) · Δ : suff. small ball in $\mathbb{T}^n$ $\Rightarrow$ $T_\gamma^{-n_k}(\Delta)\cap\Delta=\phi$

· A, B : intervals

$E_m^{-n}(A)$: $|m|^n$ intervals of length $|m|^{-n}\lambda(A)$.

uniformly spread on S^1.

(i.e. $\Delta=\left(\frac{i}{|m|^n}, \frac{i+1}{|m|^n}\right)$ intersects only one cpnt of $E_m^{-n}(A)$)

$n\uparrow$ $\Rightarrow$ B contains approximately $|m|^n\lambda(B)$ cpnts of $E_m^{-n}(A)$.

$\therefore$ $\lambda(E_m^{-n}(A)\cap B)\approx |m|^n\lambda(B)\cdot|m|^{-n}\lambda(A)=\lambda(A)\cdot\lambda(B)$

* hyperbolic toral automorphisms.

Prop. $F_L:\mathbb{T}^2\to\mathbb{T}^2$: ergodic & mixing w.r.t. Lebesgue measure.

proof) - $\chi_{m,n}(x,y)=\exp 2\pi i(mx+ny)$

$\chi_{m,n}(F_L(x,y)) = \cdots = \chi_{am+cn,\, bm+dn}(x,y)$ $\quad \left(L=\begin{pmatrix} a & b \\ c & d\end{pmatrix}\right)$

$\varphi(x,y)=\sum_{(m,n)\in\mathbb{Z}^2}\varphi_{m,n}\chi_{m,n}(x,y)$

F_L acts on $\mathbb{Z}^2$ by L^t. $\Rightarrow$ $\varphi_{m,n}=\varphi_{am+cn,\, bm+dn}$.

$\to 0$ as $m^2+n^2\to\infty$

$\therefore$ $\varphi_{m,n}=0$ $\forall (m,n)\neq(0,0)$.

· Lemma. mixing $\Leftrightarrow$ $\forall \varphi, \psi \in \Phi$: complete system of functions in L^2

$$\int \varphi(T^n x)\,\overline{\psi}(x)\,d\mu \longrightarrow \int \varphi\,d\mu \cdot \int \overline{\psi}\,d\mu \quad \text{as } n \to \infty .$$

$$\int \chi_{m,n}(F_L^N(x,y)) \cdot \overline{\chi_{k,l}(x,y)}\,d\lambda \xrightarrow{N\to\infty} \int \chi_{m,n}\,d\lambda \cdot \int \overline{\chi_{k,l}}\,d\lambda .$$

$m, n, k, l = 0 \Rightarrow 1 \to 1 \cdot 1.$

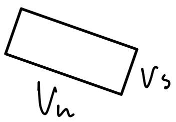
Assume $(m,n) \neq (0,0)$ $\Rightarrow \int \chi_{m,n}\,d\lambda = 0.$

$\|(L^t)^N(m,n)\| \to \infty \Rightarrow (L^t)^N(m,n) \neq (k,l)$ if $N \gg 0.$

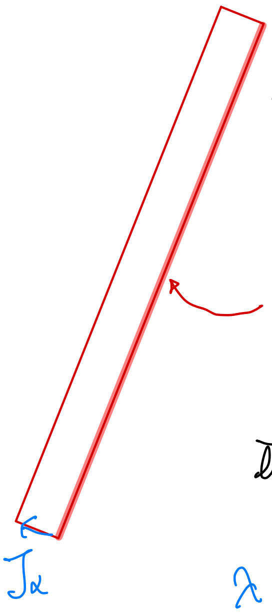
$\Downarrow$

LHS $= 0.$

geometric)



$\leftarrow$ A : parallelogram



$\leftarrow$ $F_L^{-n}(A)$: long & thin parallelogram in $\mathbb{R}^2$.

direction (v_s) linear flow : uniquely ergodic.

$J = \{T_\omega^{-t}(x)\}_{t=0}^{l_n}$

$$\frac{1}{l_n}\int_0^{l_n} \chi_B(T_\omega^{-t}(x))\,dt \to \lambda(B) \quad \text{uniformly}.$$

$$\lambda(F_L^{-n}(A) \cap B) = \int \frac{1}{l_n}\int_0^{l_n} \chi_B(T_\omega^{-t}(x_\alpha))\,dt\,d\alpha \to \lambda(A)\cdot\lambda(B).$$

*symbolic systems.

· σ_N : mixing w.r.t. μ_p (product measure of probability distribution on $0, \dots, N-1$)

$$\left(\begin{array}{l} p = (p_0, \dots, p_{N-1}), \quad 0 \leq p_i \leq 1, \quad \sum p_i = 1. \\ \mu_p(C^{n_1,\dots,n_k}_{\alpha_1,\dots,\alpha_k}) = \prod p_{\alpha_i} \end{array} \right)$$

4.3. Measure theoretic entropy.

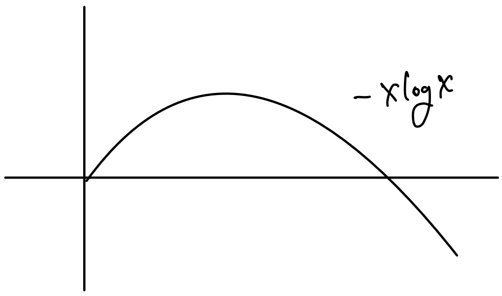
$(X, \mathcal{B}, \mu)$: probability space.

$\xi = \{ C_\alpha \in \mathcal{B} : \alpha \in I \}$: measurable partition.

if $\mu(X \setminus \bigcup C_\alpha) = 0$, $\mu(C_{\alpha_i} \cap C_{\alpha_j}) = 0$ $(\alpha_i \neq \alpha_j)$

Def. entropy of ξ ; (Set $0 \cdot \log 0 = 0$)

$$H(\xi) := H_\mu(\xi) := -\sum_{\mu(C_\alpha) > 0} \mu(C_\alpha) \log \mu(C_\alpha) \geq 0 .$$



containing x & contained in ξ.

$I_\xi : X \to \mathbb{R}$, $I_\xi(x) = -\log \mu(C_\xi(x))$: information function.

$$\Rightarrow H_\mu(\xi) = \int I_\xi \, d\mu .$$

$$0 \leq -\log\left(\sup \mu(C_\alpha)\right) \leq H_\mu(\xi) \leq \log |\xi| .$$

Def. conditional entropy of ξ w.r.t. $\eta = \{ D_\beta : \beta \in J \}$

$$H(\xi | \eta) := -\sum_\beta \mu(D_\beta) \sum_\alpha \mu(C_\alpha | D_\beta) \log \mu(C_\alpha | D_\beta) .$$

$\mu(A|B) = \mu(A \cap B) / \mu(B)$.

$$\| \quad \int I_{\xi,\eta} \, d\mu , \quad I_{\xi,\eta}(x) = -\log \mu(C_\xi(x) \mid D_\eta(x))$$

conditional information function

$$\| \quad \sum_\beta \mu(D_\beta) H(\xi_{D_\beta}) .$$

partition of D_β $= \{ D_\beta \cap C_\alpha : \alpha \in I \}$

ξ : at most countable w/ finite entropy.

Prop. · $0 \le H(\xi|\eta) \le H(\xi)$ refinement . $D\in\eta \Rightarrow \exists C\in\xi$ s.t. $D\subset C$.

· $\mu(\xi|\eta) = 0 \iff \xi \le \eta \pmod 0$

· $\mu(\xi|\eta) = H(\xi) \iff \xi,\eta$: indep. $(\mu(C\cap D) = \mu(C)\cdot\mu(D))$

· $\zeta \ge \eta \Rightarrow H(\xi|\zeta) \le H(\xi|\eta)$

· $H(\xi\vee\eta|\zeta) = H(\xi|\zeta) + H(\eta|\xi\vee\zeta)$

$$\xi\vee\eta = \{C\cap D : C\in\xi,\ D\in\eta,\ \mu(C\cap D)>0\}.$$

$$H(\xi\vee\eta) = H(\xi) + H(\eta|\xi)$$

T: measure preserving transformation, $T^{-1}(\xi) = \{T^{-1}(C_\alpha) : \alpha\in I\}$.

$\xi^T_{-n} := \xi \vee T^{-1}(\xi) \vee \cdots \vee T^{-n+1}(\xi)$: joint partition

Def. $h(T,\xi) := h_\mu(T,\xi) = \lim \frac{1}{n} H(\xi^T_{-n})$: metric entropy of T rel to ξ .

Prop. $h(T,\xi) = \lim H(\xi | T^{-1}(\xi^T_{-n}))$, $H(\xi|T^{-1}(\xi^T_{-n}))$: non-increasing.

proof) $H(T^{-1}(\xi^T_{-n}) \vee \xi) = H(T^{-1}(\xi^T_{-n})) + H(\xi|T^{-1}(\xi^T_{-n}))$.

$$\Rightarrow \underbrace{H(\xi^T_{-n+1})/n}_{h(T,\xi)} = H(\xi^T_{-n}) + H(\xi|T^{-1}(\xi^T_{-n})) .$$

$$= \cdots = \underbrace{H(\xi^T_0)/n}_{\to 0} + \sum H(\xi|T^{-1}(\xi^T_{-k}))/n$$

Def. entropy of T w.r.t. μ ;

$$h(T) := h_\mu(T) := \sup\{h_\mu(T,\xi) : \xi \text{ : measurable partition w/ } H(\xi)<\infty\}$$

Prop. $\cdot\ 0 \leq h(T,\xi) \leq H(\xi)$

$\cdot\ h(T,\xi\vee\eta) \leq h(T,\xi) + h(T,\eta)$

$\cdot\ h(T,\eta) \leq h(T,\xi) + H(\eta|\xi) \quad (\Rightarrow h(T,\eta) \leq h(T,\xi) \text{ if } \xi \leq \eta)$

$\cdot$ (Rokhlin ineq.) $|h(T,\xi) - h(T,\eta)| \leq \underbrace{H(\xi|\eta) + H(\eta|\xi)}_{d_R(\xi,\eta)\,:\ \text{Rokhlin metric.}}$

$\cdot\ h(T,\xi) = h(T, \bigvee T^{-i}(\xi))$.

Def. $\mathcal{S}$: sufficient family w.r.t. measure preserving T if
($\mathcal{S}$ ← measurable partitions w/ finite entropy.)

$\cdot$ (T : non-invertible)

partitions subordinate to $\bigvee_{i=0}^{k} T^{-i}(\xi)$ $(\xi\in\mathcal{S},\ k\in\mathbb{N})$

form a dense subset w.r.t. Rokhlin metric.

$\cdot$ (T : invertible)

same holds for $\bigvee_{i=-l}^{l} T^{i}(\xi)$ $(\xi\in\mathcal{S},\ l\in\mathbb{N})$

Rmk. $\mathcal{S} = \{\xi_n\}_{n\in\mathbb{N}}$

$\operatorname{diam}(\xi_n) \to 0 \Rightarrow \mathcal{S}$: sufficient.

($\operatorname{diam}(\xi_n)$ ← $\sup_{C\in\xi} \operatorname{diam}(C)$.)

Def. ξ : generator for T if $\mathcal{S} = \{\xi\}$: suff. family.

Thm. $h_\mu(T) = \sup_{\xi\in\mathcal{S}} h_\mu(T,\xi)$

Cor. ξ : generator for $T \Rightarrow h_\mu(T) = h_\mu(T,\xi)$.

4.4. Examples of calculation of measure theoretic entropy.

* rotations & translations.

$\xi_N = \{[\frac{i}{N}, \frac{i+1}{N}]\} \Rightarrow \{\xi_N\}$: sufficient.

$(\xi_N)^{R_\alpha}_{-n} = \bigvee_{i=0}^{n-1} R_\alpha^{-i}(\xi_N)$ has no more than $N \cdot n$ elts.

$H((\xi_N)^{R_\alpha}_{-n}) \leq \log N \cdot n = \log N + \log n$.

$\Rightarrow h_\lambda(R_\alpha, \xi_N) \leq \lim \frac{1}{n}(\log N + \log n) = 0$.

Similarly, $h_\lambda(T_\gamma, \eta = \xi_N \times \cdots \times \xi_N) = 0$.

* expanding maps.

$\xi_{|k|}$: generator for E_k

$\because \xi_{|k|,n} := \bigvee_{i=0}^{n-1} E_k^{-i}(\xi_{|k|})$: partition w/ equal intervals of length $|k|^{-n}$

entropy $= n \cdot \log |k|$

$\therefore h_\lambda(E_k) = h_\lambda(E_k, \xi_1) = \lim_{n\to\infty} \frac{1}{n} H(\xi_{|k|,n}) = \log |k|$.

* Bernoulli & Markov measures.

μ_p : Bernoulli measure $\quad (p = (p_0, \ldots, p_{N-1}), \ \Sigma p_i = 1)$

$\xi_0 = \{C^0_\alpha : \alpha = 0, \ldots, N-1\}$

$\bigvee_{i=-m}^{m} \sigma_N^i(\xi_i)$: symm. m cyl. diameter $\to 0$ as $m \to \infty$.

$h_{\mu_p}(\sigma_N) = h_{\mu_p}(\sigma_N, \xi_0) = \lim \frac{1}{n} H\left(\bigvee_{i=0}^{n-1} \sigma_N^{-i}(\xi_0)\right) = H(\xi_0) = -\Sigma p_i \log p_i$.

Cor. $h_{\mu_p}(\sigma_N) \leq h_{top}(\sigma_N) = \log N$ & equality $\Leftrightarrow p = (\frac{1}{N}, \ldots, \frac{1}{N})$

$\Pi = (\pi_{ij})$: non-negative matrix

stochastic matrix if $\sum_{i=0}^{N-1} \pi_{ij} = 1 \qquad j = 0, \dots, N-1$

$\exists$ inv. vector w/ non-negative coords. p $\quad (\Pi p = p)$.

$$\mu_{\pi,p}(C^m_\alpha) = \left(\prod_{i=-m}^{m-1} \pi_{\alpha_i, \alpha_{i+1}} \right) p_{\alpha_m} \quad : \text{Markov measure}$$

Rmk. If Π is transitive, then σ_N is mixing w.r.t. $\mu_{\pi,p}$

$$H(\xi_0 \mid \sigma_N(\xi_0) \vee \cdots \vee \sigma_N^n(\xi_0)) = - \sum_{i,j=0}^{N-1} p_j \pi_{ij} \log \pi_{ij} .$$

$$h_{\mu_{\pi,p}}(\sigma_N) = - \sum_{i,j=0}^{N-1} p_j \pi_{ij} \log \pi_{ij} .$$

$$= \sum_{j=0}^{N-1} p_j \underbrace{H(j\text{-th column})}_{-\sum_{i=0}^{N-1} \pi_{ij} \log \pi_{ij}}$$

A : transitive 0-1 matrix

(Perron-Frobenius) $\Rightarrow$ $\exists\, \lambda_{max}$, pos. eigenvector q.

v : pos. eigenvector of A^T $\longrightarrow$ $\sum q_i v_i = 1$.

$$\pi = \left(\pi_{ij} = \frac{a_{ij} v_i}{\lambda_{max} v_j} \right) \quad : \text{stochastic matrix}$$

Parry measure for TMC σ_A ; $\mu_{\pi,p}$ where $p_i = q_i v_i$.

$$h_{\mu_{\pi,p}}(\sigma_A) = - \sum q_j \cancel{v_j} \frac{a_{ij} v_i}{\lambda_{max} \cancel{v_j}} \log \frac{a_{ij} v_i}{\lambda_{max} v_j}$$

$$= \sum \frac{q_j a_{ij} v_i}{\lambda_{max}} \left(\log \lambda_{max} + \log v_j - \log a_{ij} v_i \right)$$

$$\sum q_j a_{ij} = \lambda_{max} q_j$$

$$= \log \lambda_{max} + \sum \cancel{q_j v_j \log v_j} - \sum \cancel{q_i v_i \log v_i} .$$

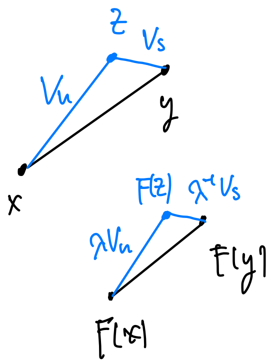
$$= \log \lambda_{max} .$$

Prop. $h_{\mu_{Parry}}(\sigma_A) = h_{top}(\sigma_A) = p(\sigma_A) = \log \lambda_{max}$.

$\ast$ $F = P_L$: hyperbolic toral automorphism. $\lambda > 1$: unstable $\Big\} \Rightarrow h_m(F) = \ln \lambda$.

ξ : partition w/ diameter $< \frac{1}{10}$

$C := \bigvee_{k=-n}^{n} F^k(\xi)$



$$d(F^k(x), F^k(z)) = \lambda^{-k} d(x,z) < \frac{\lambda^{-k}}{10}$$

$$d(F^k(x), F^k(z)) \leq \cdot + \cdot < \frac{1}{10} + \frac{\lambda^{-k}}{10} < \frac{1}{5} .$$

$$d(x,z) = \lambda^{-n} d(F^n(x), F^n(z)) < \frac{\lambda^{-n}}{5}$$

$$d(y,z) = \lambda^{-n} d(F^{-n}(y), F^{-n}(z)) < \frac{\lambda^{-n}}{5}$$

$\Downarrow$

$$d(x,y) < \frac{2}{5} \lambda^{-n}$$

$\Downarrow$

measure $\leq \frac{2\pi}{5} \lambda^{-2n}$ $\Rightarrow$ lower bound.

$$h(F, \xi) \geq \log \lambda$$

$$h_m(F) \geq \log \lambda = h_{top}(F)$$

$$h_m(F) = \frac{1}{n} F_m(\lambda^n).$$

ξ : suff. partition $\left(\frac{1}{N}\text{-squares}\right)$

$$h_m(F^n, \xi) \leq H(\xi \mid F^{-n}(\xi)).$$

$C \in F^{-n}(\xi)$: long & thin parallelogram.

diagram $< \frac{\sqrt{2}}{N} \lambda^n$

width $< \frac{1}{N} \lambda^{-n}$

$\overset{\xi}{D} \cap C \neq \phi \Rightarrow D \subset \underbrace{\frac{\sqrt{2}}{N} \text{ nbd of } C}$.

area $\leq \frac{\sqrt{2}\lambda^n + 2\sqrt{2}}{N} \cdot \frac{2\sqrt{2} + \varepsilon}{N} < \frac{10\lambda^n}{N^2}$

$\text{area}(D) = \frac{1}{N^2} \Rightarrow N(C) = \#\text{ of such } D\text{'s} < 10\lambda^n$

$$H(\xi \mid F^{-n}(\xi)) < \log \lambda^n + \log 10 = n \log \lambda + \log 10$$

$$\therefore h_m(F) = \frac{1}{n} h_m(F^n) < \log \lambda + \frac{\log 10}{n}$$

$$\Rightarrow h_m(F) \leq \log \lambda.$$

4.5. The variational principle.

Thm (Variational principle)

f : homeo on cpt (X, d)

$$h_{top}(f) = \sup_{\mu \in \mathcal{M}(f)} h_\mu(f).$$

$\mathcal{M}(f)$ ← f-inv BP measures.

$\exists$ positive entropy inv measure

$\Leftrightarrow$ positive topological entropy.

Lemma. $\mu \in \mathcal{M}$

- $x \in X$, $\delta > 0 \Rightarrow \exists \delta' \in (0, \delta)$ s.t. $\mu(\partial B(x, \delta')) = 0$.
- $\delta > 0 \Rightarrow \exists$ finite measurable partition $\xi = \{C_1, \dots, C_k\}$

$\operatorname{diam} C_i < \delta$ & $\mu(\partial \xi) = 0$.

sketch) · $\bigcup_{0<\delta'<\delta} \partial B(x, \delta')$: uncountable union of finite measure

· $B_1, \dots, B_k$: finite cover

$C_1 := \overline{B_1}$, $C_i := \overline{B_i} \setminus \overline{B_1} \cup \dots \cup \overline{B_{i-1}}$

Lemma. $E_n \subset X$: (n, ε)-separated set

$\nu_n := \frac{1}{\operatorname{card} E_n} \sum_{x \in E_n} \delta_x$: uniform δ measure on E_n

$\mu_n := \frac{1}{n} \sum_{i=0}^{n-1} f_*^i \nu_n$

$\Rightarrow \exists$ accumulation pt μ of $\{\mu_n\}$ (in the weak* top) s.t. f-inv. &

$$\limsup_{n \to \infty} \frac{1}{n} \log \operatorname{card} E_n \leq h_\mu(f)$$

Separated set $\Rightarrow$ entropy ↑ measure.

sketch) $\xi = \{C_1, \dots, C_k\}$: partition of X.

μ : Borel $\Rightarrow$ $\mu(C_i) = \sup\{\mu(B) : B \subset C_i : \text{closed}\}$.

Choose $B_i \subset C_i$ s.t. $\beta = \{\underset{\text{sea}}{B_0}, \underbrace{\dots, B_k}_{\text{islan}}\}$ $(B_0 = X \setminus \cup B_i)$

$H(\xi | \beta) < 1$.

$$h_\mu(f, \xi) \leq h_\mu(f, \beta) + H_\mu(\xi | \beta) \leq h_\mu(f, \beta) + 1$$

$\mathcal{B} := \{B_0 \cup B_1, \dots, B_0 \cup B_k\}$: open cover of X.

$H_\mu(\beta^f_n) \leq \log 2^n \text{card } \beta^f_n$

δ_0 : Lebesgue # of $\mathcal{B}$.

" of $\mathcal{B}^f_{-n}$ w.r.t. d^f_n.

$\Downarrow$

δ_0 - separated set

$$h_\mu(f, \xi) \leq h_\mu(f, \beta) + 1 \leq h_{top}(f) + \log 2 + 1.$$

$$h_\mu(f) = \frac{h_\mu(f^n)}{n} \leq \frac{h_{top}(f^n) + \log 2 + 1}{n}$$

$$\therefore h_\mu(f) \leq h_{top}(f)$$

Q. $\exists \mu$ s.t. $h_\mu(f) = h_{top}(f)$?

Thm. f: expansive $\Rightarrow$ $\exists$ such a measure μ.

(measure of maximal entropy).